MIL-H-52544B(ME)
23 September 1981
SUPERSEDING
MIL-H-52544A(ME)
17 December 1974

# MILITARY SPECIFICATION

## HOSE, RUBBER; OIL SUCTION, WIRE-REINFORCED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers wire-reinforced hydraulic suction hose.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-P-775 | - Packaging of Hose, Hose Assemblies; Rubber, Plastic, Fabric, or Metal (Including Tubing); and Fittings, Nozzles, and Strainers. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4720

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STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-1188 | - Commercial Packaging of Supplies and Equipment. |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The hose shall be in accordance with Figure 1 and as specified herein.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.3.1 Tube. The tube of the hose shall be seamless oil-resistant synthetic rubber.

3.3.2 Cover. The cover shall be seamless ozone- and oil-resistant synthetic rubber.



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3.4 Low temperature. The hose, when attached to adapters and while at minus 40° C (minus 40° F), shall withstand bending to the minimum bend radius specified in figure 1 within 8 to 12 seconds without evidence of chipping or cracking.

3.5 Reinforcement. The reinforcement of the hose shall be a ply or plies of woven or braided textile fibers with a spiral or spirals of steel wire. Each individual wire shall be continuous and shall contain no splices. The individual loops of the spirals shall be evenly spaced. When the reinforcement contains two or more spiral wires, the individual wires shall not touch or cross over each other.

3.6 Ozone resistance. The hose cover shall withstand 50 parts of ozone per 100 million parts of air maintained at plus 38° C (plus 100° F) for 70 hours without evidence of cracking when observed under 7-power magnification.

3.7 Oil resistance. After immersion in oil at a temperature of plus 100° C (plus 212° F) for 70 hours, the change in volume of butadiene acrylonitrile shall not exceed 30 percent, the change in volume of polymerized chloroprene shall not exceed 100 percent, and there shall be no shrinkage.

3.8 Maintenance of vacuum. While attached to adapters, the hose shall withstand an internal vacuum of 635 millimeters (25 inches) of mercury without evidence of separation of the tube from the body of the hose, collapse of the tube, a decrease in the outside diameter exceeding 15 percent, or leakage.

3.9 Proof pressure. While attached to adapters, the hose shall withstand pressurization equal to twice the applicable working pressure specified in figure 1 without evidence of leakage, rupture or detachment (see 6.4).

3.10 Burst pressure. While attached to adapters, the hose shall withstand pressurization equal to four times the applicable working pressure specified in figure 1 without evidence of leakage, rupture or detachment.

3.11 Age. The hose shall be manufactured not more than twenty calendar quarters (5 years) prior to date of delivery to the Government.

3.12 Length. Unless otherwise specified (see 6.2), bulk hose shall be furnished in lengths of 13.7 meters (M) (45 feet) or longer except that not more than 25 percent may be furnished in lengths between 7.6 M (25 feet) and 13.4 M (44 feet) and not more than an additional 25 percent may be furnished in lengths between 4.6 M (15 feet) and 7.3 M (24 feet).

3.13 Size. The size of the hose shall be selected from figure 1, as specified (see 6.2).

3.14 Marking. The hose shall be marked on the layline. Order of marking may be in any sequence. The marking shall include but shall not be limited to the following:



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(a) R4 or 100R4.
(b) Nominal size - 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3, 3-1/2, or 4-inches.
(c) Hose manufacturer's name, trademark or code.
(d) Date of manufacture (quarter and year).

3.15 Workmanship. The hose cover shall contain no patch or blister and shall be free from wrinkles, except that minor impressions less than 0.80 mm (0.031-inch) deep (left by the curing wrap) will be acceptable. The hose cover and tube shall contain no laps, laminations, cracks, or holes and shall show no evidence of looseness (wrinkles when bent). The hose tube shall show no evidence of ridges. There shall be no evidence of reinforcement wire through the hose tube or cover. The hose shall be cleaned free from oil, grease, dirt, or other foreign material, both internally and externally. Mandrel lubricants not readily removable are allowed provided they are not detrimental to hydraulic system components and fluids.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The contractor is responsible for insuring that materials used are manufactured, examined, and tested in accordance with reference specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article hose shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the first article hose shall be tested as specified in table I. Unless otherwise specified herein, the tests shall be conducted in the sequence listed in table I. Failure of any test shall be cause for rejection.



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TABLE I. Test schedule.

| Schedule | | Designation | Test paragraph | Number of specimens per sample |
|---|---|---|---|---|
| Test series | Sequence | | | |
| A | 1 | Low Temperature. | 4.5.2.2(a) | 2 |
| | 2 | Vacuum collapse. | 4.5.2.2(f) | |
| | 3 | Proof pressure. | 4.5.2.2(d) | |
| B | 1 | Vacuum collapse. | 4.5.2.2(f) | |
| | 2 | Burst pressure. | 4.5.2.2(e) | 2 |
| C | 1 | Ozone resistance. | 4.5.2.2(b) (see 6.5) | 3 covers |
| D | 1 | Oil resistance. | 4.5.2.2(c) | 3 covers 3 tubes |

4.4 Quality conformance inspection.

4.4.1 Lot size. A lot shall consist of not more than 914.4 m (3000 feet) of hose produced over a period not to exceed 30 days. Each lot shall contain hose of one nominal size, manufactured under identical conditions, by one manufacturer in one plant.

4.4.2 Unit of product. For examination sampling purposes, a unit of product shall be defined as 609.6 mm (2 feet) of hose.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4.5 Tests.

4.4.5.1 Individual. Each length (see 3.12) of hose in the lot shall be tested in accordance with 4.5.2.2(d). Failure of the test by any length of hose shall be cause for rejection of that length of hose.



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4.5 Inspection procedure.

4.5.1 Examination. The hose shall be examined as specified herein for the following defects:

Major

101. Hose does not conform to dimensions and tolerances specified in figure 1.
102. Materiel not as specified.
103. Reinforcement not as specified.
104. Age not as specified.
105. Length not as specified.
106. Size not as specified.
107. Workmanship not as specified.

Minor

201. Marking not as specified.
202. Depressed area of cover (exceeds outside diameter minimum tolerance).
203. Depressed area of hose tube (exceeds inside diameter maximum tolerance).
204. Presence of foreign material.

4.5.2 Tests.

4.5.2.1 Specimen preparation. Specimens shall be fabricated into test assemblies in accordance with SAE J343.

4.5.2.2 Hose. The hose shall be tested in accordance with SAE J343 as follows:

| Test | Requirement paragraph |
|---|---|
| (a) Low temperature | 3.4 |
| (b) Ozone resistance | 3.6 |
| (c) Oil resistance | 3.7 |
| (d) Proof pressure | 3.9 |
| (e) Burst pressure | 3.10 |
| (f) Vacuum collapse | 3.8 |

Nonconformance to the applicable requirements shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for level A and level B shall be examined to determine compliance with the applicable quality assurance provisions of MIL-P-775. The preservation, packing, and marking for commercial shall be examined for compliance to MIL-STD-1188.



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5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Hose shall be preserved in accordance with the level A requirements of MIL-P-775.

5.1.2 Commercial. Hose shall be preserved in accordance with MIL-STD-1188, except that unit pack is not required.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Hose shall be packed in accordance with the level A requirements of MIL-P-775.

5.2.2 Level B. Hose shall be packed in accordance with the level B requirements of MIL-P-775.

5.2.3 Commercial. Hose shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military packaging shall be in accordance MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for use in hydraulic suction lines of hydraulic systems on equipment within the temperature range of minus 40° C to plus 121° C (minus 40° F to plus 250° F).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is required for inspection and approval and number of units required (see 3.2, 4.3 and 6.3).
(c) Length of hose, when a specific length is required (see 3.12).
(d) Size of hose required (see 3.13).
(e) Degree of preservation and degree of packing required (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation (DAR). The first article should be a first produced hose. The contracting officer should include specific instructions in all procurement instructions, regarding arrangements for examinations, test and approval of the first article.



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6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 Leakage. Any passage of fluid from the inner postion of an adapter, hose, or hose assembly, as determined by sight, touch, or by pressure loss. Leaks occur through the hose, the adapter body, at the junction between the hose and adapter or at the sealing surface (thread or flange face) of an adapter. Failure of a test fixture is not included.

6.4.2 Rupture. A leak which causes visible damage to the adapter or hose, as evidence by the rapid loss of volume of the pressurizing fluid, or sharp reduction in pressure.

6.4.3 Detachment. The loss of contact between the adapter and the hose to which it is attached, or the loss or partial loss of contact between the adapter and hose or the adapter and test fixture fitting by virtue of thread stripping or severance of an adapter body. Failure of a test fixture fitting is not included.

6.5 Ozone resistance test. It is recommended that the procuring activity accept certification in lieu of ozone testing when the contractor can provide records showing that the ozone resistance test has been run for the Government within the past 2 years [see 3.6 and 4.5.2.2(b)].

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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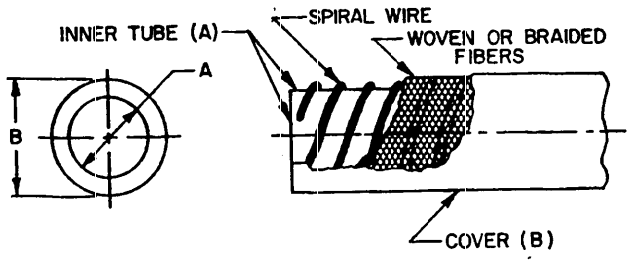
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| NOMINAL HOSE SIZE I.D. | A INNER TUBE | | | | B COVER OUTSIDE DIAMETER MAX. | | MINIMUM BEND RADIUS | | MAXIMUM WORKING PRESSURE | |
|---|---|---|---|---|---|---|---|---|---|---|
| | INSIDE DIAMETER | | I.D. TOLERANCE | | | | | | | |
| | MM | IN | MM | IN | MM | IN | MM | IN | MPA | PSI |
| 3/4 | 19.0 | .750 | ± 0.79 | ±.031 | 34.9 | 1.375 | 127.0 | 5 | 2.07 | 300 |
| 1 | 25.4 | 1.000 | ± 0.79 | ±.031 | 41.3 | 1.625 | 152.4 | 6 | 1.72 | 250 |
| 1-1/4 | 31.8 | 1.250 | ± 1.19 | ±.047 | 50.8 | 2.000 | 203.2 | 8 | 1.38 | 200 |
| 1-1/2 | 38.1 | 1.500 | ± 1.19 | ±.047 | 57.2 | 2.250 | 254.0 | 10 | 1.03 | 150 |
| 2 | 50.8 | 2.000 | ± 1.57 | ±.062 | 69.9 | 2.750 | 304.8 | 12 | 0.69 | 100 |
| 2-1/2 | 63.5 | 2.500 | ± 1.57 | ±.062 | 82.6 | 3.250 | 355.6 | 14 | 0.43 | 62 |
| 3 | 76.2 | 3.000 | ± 1.57 | ±.062 | 95.3 | 3.750 | 457.2 | 18 | 0.39 | 56 |
| 3-1/2 | 88.9 | 3.500 | ± 1.57 | ±.062 | 107.9 | 4.250 | 533.4 | 21 | 0.31 | 45 |
| 4 | 101.6 | 4.000 | ± 1.57 | ±.062 | 120.7 | 4.750 | 609.6 | 24 | 0.24 | 35 |

NOTES:
1. DIMENSIONS ARE IN MILLIMETERS (MM) AND INCHES (IN).
2. PRESSURES ARE IN MEGAPASCALS (MPA) AND POUNDS PER SQUARE INCH (PSI).
3. INSIDE DIAMETERS ARE MEASURED AT LEAST 50.8 MM (2 INCHES) IN FROM THE END.

FIGURE 1. Hose, rubber, oil suction, wire-reinforced.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-H-52544B(ME) Hose, Rubber; Oil Suction, Wire-Reinforced

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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